

Sprint 5

Sprint Goal

Develop the core React frontend by implementing the operational dashboard, container management interface and end-to-end approval workflow while integrating the frontend with the existing API

Sprint Backlog

ID	User Story	Priority	Story Points	Status
PB-18	Develop dashboard interface	Must Have	5	Done
PB-19	Implement dashboard and summary cards	Must Have	5	Done
PB-20	Develop a container details page	Must Have	5	Done
PB-21	Complete Frontend approval workflow	Must Have	8	Done

Development Tasks

- Develop the React dashboard interface
- Display live operational summary statistics
- Integrate the dashboard with the backend REST API
- Develop the container details page
- Display lifecycle information for each container
- Complete the end-to-end container approval workflow
- Validate frontend integration with JWT authentication
- Test dashboard functionality and navigation

GitHub Evidence

- 22 July 2026- Implement responsive dashboard with live summary data
- 22 July 2026- Complete Sprint 5 dashboard implementation
- 22 July 2026 – Added container details page with lifecycle information
- 23 July – Complete end-to-end container lifecycle approval workflow on frontend

Definition of Done

- Dashboard interface operational
- Live operational statistics displayed
- Container Details page implemented
- Frontend approval workflow completed
- Backend API integrated successfully
- JWT authentication validated
- Features tested and committed to GitHub
- Sprint deliverables integrated successfully

Sprint Review

Sprint 5 delivered the core React frontend for the Container Management System. A responsive operational dashboard was developed and integrated with the backend API to display live summary statistics. The Container Details page was implemented to present lifecycle information for individual containers, while the complete end-to-end approval workflow was successfully integrated into the front end. By the end of the sprint, users could interact with the system through a functional web interface backed by the previously developed REST API.

What Went Well

- Dashboard integration with the backend was completed successfully
- Live operational statistics were displayed using API data
- Container lifecycle information was presented through a Container details page
- Frontend approval workflow aligned with backend business rules

Challenges

- Integration frontend components with authenticated API endpoints
- Managing the application state across multiple pages
- Designing responsive layouts using Material UI
- Ensuring frontend behaviour remained consistent with backend workflow validation

Focus for Sprint 6

- Implement search functionality
- Complete approval management pages
- Enhance dashboard charts and operational reporting
- Perform end-to-end testing on the system
- Complete regression analysis
- Finalise project documentation